

Manual

Collection / Information Functions for Material, ERP Batches, MES Batches AIP-MTR 8.2

Version 1.0.23049

Last changed on: 01.09.2020

Copyright

©Copyright 2020 All rights reserved.

SAP® and R/3® are registered trademarks of SAP AG.

WINDOWS® is a registered trademark of Microsoft Corporation.

MPDV® and HYDRA® are registered trademarks of MPDV Mikrolab GmbH.

ORACLE® is a registered trademark of ORACLE Corporation, California, USA.

Copying and distribution of this documentation or any part thereof, for any purpose or in any form, is prohibited without prior written permission from MPDV Mikrolab GmbH.

The information contained in this documentation is subject to change without prior notice.

Contents

1	Overview of Collection / Information Functions for Material and Batches	6
2	AIP2 Operation.....	8
2.1	Special Control and Display Elements on the AIP2	8
2.2	General description of the posting process on the AIP2	11
3	Main View with Tiles.....	15
3.1	Main view – header and footer	15
3.2	Main view with "tiles"	16
3.3	Icon view of workplaces	23
4	Basic Screen as List View.....	25
4.1	Basic screens – header and footer	25
4.2	Basic screen "tabular view"	27
4.3	Basic screen "machine overview"	30
4.4	"Machines as icons" basic display	33
5	Batch-related Entry Function.....	34
5.1	Basic screen.....	34
5.2	Order postings for operations subject to batch management.....	34
5.3	Batch-related postings.....	37
5.4	Run-through batch mode.....	45
5.5	Manual report batch quantity in the MPL environment.....	45
5.6	Display produced output batches	46
5.7	Batch information	48
5.8	Display of consumption balance	48
5.9	Advance logon of input batches.....	49
	General / usage.....	49
	Configuration.....	50
	Procedure	50
	Usage at AIP	50

5.10	Collection of serial numbers	52
5.10.1	Entry of serial numbers for OPs that are not subject to management in batches (dialog A_SNR/ "E")	52
5.10.2	Entry of serial numbers for OPs that are subject to management in batches (dialog A_SNR)	55
6	Barcode Input with Prefix	59
6.1	Configuration of customized barcode prefixes	63
7	AIP2 -Local Configuration	65
7.1	Local Configuration ctaip.ini	65
7.2	PNG – Files / Bitmaps	69
7.2.1	File pict.zip	69
7.2.2	File pict_cust.zip	69
7.3	Multilingualism (*.mld files)	70
8	AIP2 - Central Configuration File hytnrcfg.ini	71
8.1	Layout configuration	74

1 Overview of Collection / Information Functions for Material and Batches

Purpose

The AIP features contained in this function package make it possible to enter batch related data directly in production using shop floor terminals or data entry PCs.

Integration

Data entered using AIP can be displayed in various applications or evaluated in MOC. The entered data can be uploaded via relevant interfaces.

Features

Order-related data entry and posting functions

- Batches can be entered in parallel to operation postings
- Entry and validation check of input and output batch change while the operation is running
- Entry of goods receipt batches and automatic generation of goods receipts/ goods issues
- Batch numbers can be entered at the same time that operations are logged on (configurable per operation and workplace)
- A ticket/ label is automatically printed in HYDRA standard format using the assigned printer when a new batch is generated
- A validation check as to whether documentation is required is run for batches when the operation is logged on
- Entry of batch changes during order processing
- Entry of batch numbers via keyboard and/ or barcode.
- Batch-related quantity and time input

Functions for entering/ generating and displaying series and serial numbers:

- Predefined series or serial numbers are transferred via the HYDRA
- ERP interface in HYDRA standard format
- Serial numbers are entered or generated via a user-friendly posting function at the BDE terminal
- Possibility to assign components that can also be identified by serial numbers
- Classification into yield/ scrap quantities with reasons
- Automatically generate goods receipts/ goods issues
- Optional verification of serial numbers that were already defined to the order

- Entry of order-related series and serial numbers to identify the processed parts
- Validation check for already posted series or serial numbers
- Display available and already used series or serial numbers at workplace PCs (MES Operation Center)
- Upload of the posted series and serial numbers via the HYDRA ERP interface in HYDRA standard format

Additional licenses may be needed in order to use the functions listed above. Adding and coordinating the specific requirements and implementing them are considered a customized HYDRA service.

2 AIP2 Operation

2.1 Special Control and Display Elements on the AIP2

Tables

Uniform selection lists are used in AIP 8.2 posting dialogs:

Filter	
Status	Status
1	PRODUCTION
2	SETUP
3	ORDER SHORTAGE
4	OUT OF MATERIAL
5	OUT OF STAFF
6	START UP
22	COLOUR CHANGE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE
99	GENERAL DISTURBANCE

Page 1 • Page 2

If information is available for more than one page, the page numbers are displayed below the table. The current page is highlighted in bold letters. If the user clicks/touches a page, the display directly changes to this page.

If more pages are available than the page numbers displayed, the following buttons can be displayed on the left or right hand side depending on the context (available as of SP10/2016):

-  : If you click this button, the system jumps to the first page of the next page navigation. This means: If *Page 1 ... Page 9* were displayed for the page navigation, the system jumps to *Page 10*.
-  : If you click this button, the system jumps to the first page of the next page navigation. This means: If *Page 10 ... Page 18* were displayed for the page navigation, the system jumps to *Page 9*.
-  : If you click this button, the system directly jumps to *Page 1*.
-  : If you click this button, the system directly jumps to the last page.

You can select an operation using the mouse, touch screen, keyboard (arrow keys: '↑' or '↓'), scanner or by entering it manually.

The content of tables or lists depends on the respective context. Example: When you log on an operation, those operations are available that are included in the sequencing list or planned for the respective workplace or group. When you interrupt an operation, only running operations are available for selection.



Scrolling page by page (up or down) in the table.



Scrolling to the left or right. Only those buttons are activated that make sense for the current situation (context sensitive). This figure shows that scrolling to the left has been deactivated.

Filter

Optionally you can display a “table filter” (customization). This is an automatic filter that, once it has been entered, directly affects the table without having to update it. This process is realized through full-text search for (defined) columns. The search is *case-insensitive*.

Virtual keyboard

Using the virtual keyboard, you can enter data manually via touch screen or a connected mouse. The virtual keyboard is displayed automatically as soon as the focus is on an input field. The keyboard layout, which is installed and activated in the Windows language settings, specifies the layout of the virtual keyboard.



Moving the virtual keyboard



Hiding the keyboard for 10 seconds



Switching between the alphanumeric and numeric keyboard



Selecting the keyboard layout (language)



Changing the scaling/size of the keyboard



To move the keyboard, you must configure the driver accordingly (configuration in the control panel of the terminal/PC)!

If you do not want to display the virtual keyboard in general, you must enter the parameter `-t` in the entry `parameters=` of the configuration file `ctaip.ini`.

Date display

AIP supports a country-specific date format in dynamic dialogs. The option "short date" has to be selected in the "regional settings" of the Windows "control panel" of the terminal/PC. Please note:

- Years are always four characters long.
- Months and days are always 2 characters long.
- Allowed separators are: '-' (minus), '/' (backslash) and '.' (dot).
- Blanks must not be included in the "short date" format, i.e. the <BLANK> separator is not allowed.
- The date separator "." (dot) is only allowed in connection with the DD.MM.YYYY format.
- The date format, which might possibly be configured in dynamic dialogs, is ignored.

Examples

- | | |
|-----------------------|------------|
| • English(USA) | MM/DD/YYYY |
| • Danish | MM-DD-YYYY |
| • Customer-specific 1 | YYYY-MM-DD |
| • Customer-specific 2 | YYYY/MM/DD |

- Customer-specific 3 MM/YYYY/DD

Note

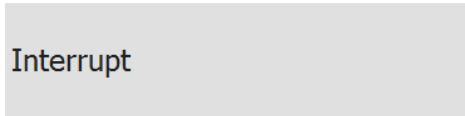
If the date format used is other than the permitted formats, a note appears when the program is started and the date format is set to MM/DD/YYYY.

In the status bar, the year format is shortened and displayed only with two characters.

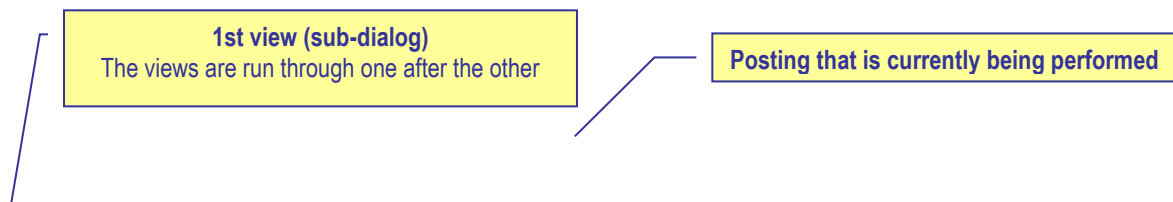
2.2 General description of the posting process on the AIP2

Many AIP posting dialogs are divided into several views (sub-dialogs). These views (sub-dialogs) cover the entire screen so that only one dialog is visible at a time. In a “workflow concept” the user is navigated through the posting dialog step by step. In the following, this process is described using the example *Interrupt operation*. Other posting dialogs are operated in the same way.

The action *Interrupt operation* is performed. To start this action, you click the button *Interrupt* when you have selected an operation:



The dialog *Interrupt operation* opens and the first view (sub-dialog) is displayed. The header displays the function that is currently being executed (here: *Interrupt operation*).



In the first dialog *Enter quantities*, the user can enter the produced yield or scrap quantities. Subject to the active input field, the virtual keyboard is shown or hidden automatically.

Quantities can be entered using the virtual or real keyboard. The user can go to the next field using the tabulator key (which can also be found on the virtual keyboard). When the user has entered all values in the first view, the next view (sub-dialog) can be opened by clicking **Next**.

The **Cancel** button is displayed in all sub-dialogs. Click this button to cancel/close the entire process at any time.

To open the next view (**Select status** in the example), click the **Next** button or another tab (in our example: **Select status** or **Confirmation**). Please note in this context, that no view can be skipped when they are navigated upwards (view 1 → view 2 → view 3). This means: When you are in the first view (*enter quantities*) and you click the third view (*confirmation*), the second view (*select status*) will be displayed first.

Vice versa, when navigating downwards (e.g. from the *confirm* view to the *enter quantities* view), each view can directly be opened by clicking the required tab. In this case, views can actually be skipped. Using the **Back** button, views are opened one after the other (upwards).

As long as the dialog has not been confirmed, the data entered can be changed at any time by scrolling back and forth.

Interrupt operation S61003050100 InjectionMolding at workplace 60610

1 Enter quantities 2 Choose status 3 Confirmation

Workplace 60610 ENG21

Actual status OUT OF MATERIAL since 02.10.2015 15:40:10

Filter

Status	Status
1	PRODUCTION
2	SETUP
3	ORDER SHORTAGE
4	OUT OF MATERIAL
5	OUT OF STAFF
6	START UP
15	MAINTENANCE
20	REPAIR
22	COLOUR CHANGE
24	TECHNICAL FAILURE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE

Page 1 • Page 2

New status 4 OUT OF MATERIAL Estimated duration 0 Minutes

Comment

Cancel Back Go on

04.12.15 15:15:00

In the second view *Select status*, you select the workplace status that is set, when the operation has been interrupted. You can select the status from the status list displayed. This list can be restricted using the *Filter* field. Once the required values have been entered, the next view/sub-dialog can be opened by clicking **Next** (in our example it is the last view).

Interrupt operation S61003050100 InjectionMolding at workplace 60610

1 Enter quantities 2 Choose status 3 Confirmation

Operation 0100 InjectionMolding

Prod. Order	S6100305	Target qty.	90000 PCS
Article	SAR-FM0011-K-5013	Yield	1442 PCS
	Mirror housing right, blu	Scrap	33 PCS
		Completion	1%

Workplace 60610 ENG21

New status OUT OF MATERIAL

Entered quantities

Yield	5
Scrap	0

Staff Number

Buttons: Cancel, Back, Interrupt OP

Status bar: 04.12.15 15:15:45

The sub-dialog *Confirmation* shows a summary of all values entered in the dialog. If the user agrees with the entered data, the **Interrupt operation** dialog can be confirmed, once the badge number has been entered. Then the dialog is sent to the server and posted.

If an input field of the dialog is not completed properly (e.g. a mandatory field is empty), the field is highlighted in red in the respective view and gets the focus. The user can then directly correct the value.



If a workflow dialog is opened, you can click the ESC key to directly exit the dialog. This exit is also possible, if the virtual keyboard is displayed. As a consequence, you cannot use the ESC key to close the virtual keyboard.

3 Main View with Tiles

With the AIP2, the user can switch between the tile design optimized for touchscreens and the list format. By default, the tile layout is shown, which is described in the sections that follow.



To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

3.1 Main view – header and footer

Header



The AIP logo is displayed on the top left of the screen, which may be replaced with a customer logo after configuration.

Possible messages are displayed to the right of it (e.g. if a dialog is opened for more than five minutes).

A separate window opens to display error messages that occur during data collection (e.g. validity checks).

Main views

You can assign a maximum of 16 workplaces or machines to the AIP2 terminal. The different workplaces are listed in the order that they were assigned to the terminal on the client. .

In the main view of the AIP2, you can use the button "< Overview" to switch to the icon view of workplaces. In the terminal configuration of the client you can specify whether you want to use the icon view. The sections that follow describe the main view and the icon view.

Footer



The MPDV logo can be found at the bottom left of the AIP2 terminal. Double clicking the logo opens the info dialog where you can start further administration functions. This dialog closes automatically after approx. 5 seconds.

In the middle, further information is displayed: the current terminal status, AIP2 version number, date of the build, IP address of the server and the terminal number.

The current date and time are displayed to the right.

AIP2 statuses

	Network connection has been established. The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.
	The terminal is sending data to the server.
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information, are disabled. But you can still record certain postings. These postings are transferred to the server, once data connection has been re-established.
	Data is being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending saved data records to the server.
	DEMO mode The terminal is in DEMO mode, i.e. server communication is disabled.

3.2 Main view with "tiles"

The screenshot displays the AIP2 main view interface. The interface is organized into several functional areas:

- Left Sidebar:** Contains a 'List of workplaces' tile showing a list of production units with their respective status and counts.
- Main Content Area:**
 - Workplace/machine:** A tile showing details for a specific machine, including its short name, group, status, start time, duration, target cycle, actual cycle, yield, and scrap.
 - Operations:** A tile showing details for a specific operation, including the MES order number, article, and quantities.
- Bottom Section:**
 - Staff logged on:** A tile showing the names and IDs of staff members currently logged on.
 - Staff tiles/Resource tiles/Material tiles:** A tile showing additional information related to staff, resources, and materials.

Please note: The actual display can be different to the above illustration.

Subject to the configurations made, the main view with tiles consists of two or three rows of tiles. While the first two rows of tiles (workplace and operation tiles) are always displayed, it is up to the user whether or not the third row of tiles is shown (optional display). In the configuration of workplaces you can configure for each workplace separately if you want to show the third row of tiles.

Additionally, there is the list of workplaces to the left. Here, you can select the workplaces for which details are displayed on the right-hand side.

List of workplaces

The workplace list shows all workplaces/machines assigned to the terminal. If many workplaces are assigned, swipe to get to the workplaces displayed further down.

This information is shown for the workplaces:

Machine/workplace number

Shows the machine and/or workplace number.

Status

The status is displayed for each machine in color on the left-hand side and also the status text is colored.

Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

If the production lock is enabled, an exclamation mark is displayed in the same color as the status.

Quantities

On the right-hand side, the first figure shows the produced yield in green and the red figure shows the produced scrap.



If you have enabled the *Compensate manual quantities* option (e.g. set off scrap against yield) and the machine list also shows shift-related quantities, they will not be updated immediately. The application only updates the quantities, once the lists have been reloaded.

Unit for yield and scrap

If no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on, the primary quantity unit of the operation is displayed.

Workplace tiles

The workplace tiles provide the following details:

Workplace/machine

No workplace or machine number is displayed.

Short name / group

The short name of the workplace or machine and the machine group are displayed.

Status

The status is displayed in color and as status text on the left-hand side. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

If the production lock is enabled, an exclamation mark is displayed in the same color as the status.

Machine image

Shows the picture of the machine stored in the configuration of workplaces.

Clocks

Shows the recorded machine cycles of the current shift.

Start / Duration [hrs:min]

Point in time since the status has been available and the resulting duration at the current point in time.

With BDE workplaces¹, this point in time refers to the last manual status change. With MDE workplaces, it refers to the time when the last status change was identified (for machine connections). It also refers to the point in time when the status was last changed manually or to the time of the last shift change.

Target/actual cycle

Current target and actual cycle of the workplace.

The largest target cycle of all operations logged on to the workplace is shown. The largest target cycle is transferred to the MDE for monitoring.

If the target cycle is smaller than the minimum cycle time, the target cycle is still shown.

¹ An MDE workplace is a workplace that is assigned to a terminal, which runs in the "MDE" operation mode. Otherwise, it is a BDE workplace.

If an operation is logged off or interrupted, the largest target cycle of the remaining operations is identified and displayed. After logoff or interruption of the last operation at the workplace, the last target cycle set is still displayed.

If no operation is logged on, the target cycle specified in the machine list is displayed. Thus, even after a restart, the terminal can get the target cycle that last applied.

Yield / Scrap

Yield and scrap quantities of the current shift produced at the machine/workplace.

KPIs: OEE, utilization efficiency, scrap ratio



This function is only available if you enable the extension [aipkpi](#).

Shows the KPIs OEE, utilization efficiency and scrap ratio. The application calculates the KPIs at cyclic intervals (scheduler job "MDE keyfigure calculation"). The KPIs always refer to the current shift. The application calculates the KPIs based on the formulas that are also used for the OEE report and/or the efficiency report on the MOC. The application shows the KPIs with two decimal places. To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached. If you have not defined limit values, the application shows the KPI in gray, otherwise in the color you defined for exceeding/not reaching limit values. For further information on the configuration, refer to the document [MDE KPI Configuration.pdf](#).



The AIP calculates and updates data at cyclic intervals. This may result in deviations between the collected values and the displayed KPIs.

Linked functions

If you click the workplace status, the dialog for changing the status opens. If you click one of the other tiles, the dialog opens where you can start the functions available for the selected workplace.



The buttons displayed depend on the selected workplace. The button *Lock production status*, for example, is only available for MDE machines.

List of operations logged on

The middle area on the right-hand side shows the logged on operations as tiles. The following data is shown:

MES order number

Order number and operation number of the operation logged on. The combination of these two numbers is the *MES order number*.

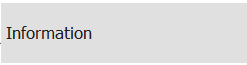
Article

Article defined for the operation.

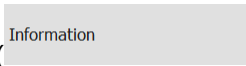
Quantities (target / yield / scrap)

Shows the target quantity defined for the operation, the produced yield and the scrap. The yield and scrap quantities integrate the counter readings of the available machine connections.



This icon is displayed if at least one note has been recorded for this operation in the graphic planning board of the client that must be shown on the terminal. To display the note(s), click the icon or click the operation and select the *Information* button ().



This icon is displayed if at least one long text is stored for this operation. Click this icon to display the long texts or click the operation and select the button *Information* ().

If you click an operation, a screen opens that shows the workplace/operation data. Via this screen, you can also select the operation-related functions.

Operation tiles

In addition to the fields already described, the operation tiles also show the following data:

Comments

This tile shows the user fields 53 and 54 (alphanumeric, 20 characters) of the operation. To edit these fields, you must store a respective user field key for the operation, which includes these two fields.

Completion in %

The bar shows the proportion of "yield", which has been produced until now, compared to the "target quantity".

Since logon (target / yield / deviation)

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time when no production lock has been set for the machine). If the terminal program has been restarted after the OP logon, no value can be calculated.

Calculation:

Target Since Logon = Net Running Time[sec] * Partitioning/Target Cycle[sec/stroke]

Net Running Time: Time since logon while the production lock has not been set. This calculation does not integrate the breaks specified in the shift model or the status times posted to RPA 12 (resource performance account).

Deviation (in percent) between the calculated target quantity since logon and the quantity which has actually been produced "since logon".

Calculation: Deviation[%] = 100% * (Yield Since Logon - Target Quantity Since Logon) / Target Quantity Since Logon.

As of CTAIP 8.2.1.32:

Up to now, the target quantity since OP logon was only calculated if the workplace/machine was assigned to a terminal with operation mode "MDE processing". As of CTAIP 8.2.1.32, the target quantity since OP logon is also calculated if the terminal is configured with operation mode "BDE processing".

If the terminal has been restarted after an operation logon, the target quantity cannot be calculated correctly. To improve transparency, an "*" (asterisk) is shown behind the target quantity (since OP logon) in this case. The asterisk indicates that the target quantity now displayed no longer refers to the time of the operation logon, but to the time of the terminal restart.

While the workplace/machine status 999 is displayed, the target quantity since OP logon is "---". If the status 999 is again changed within a "free shift", the target quantity since OP logon is calculated using the point in time of the OP logon, of the shift start or of the terminal start.

To disable the calculation of the target quantity since OP logon on the terminal, you can use the configuration CalcTargetYieldSinceLogon=0 in the hytnrcfg.ini. The quantity is then displayed using "---".



With workplaces/machines that are configured as "Machining centers", the calculation of the target quantity since OP logon is generally disabled because this calculation contradicts the principles of the machining center.

Planned duration

The field *Planned duration* displays the target processing time of the operation in format [h:mm].

Partitioning

Calculate the displayed partitioning as follows:

TLG _M	Partitioning of the workplace/machine (TLG in mnr.lst)
DIV _M	Pulse factor of workplace/machine (IMPFAKT in mnr.lst)
TLG _{AG}	Partitioning of the operation (TLG in anr.lst)

The application shows the calculated partitioning without decimal places, provided it is an integer value.

In case the partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Displaying the "3rd list"

The third list is optional. You can configure the third list in the configuration of workplaces. The following lists can be displayed:

- List of staff logged on to the currently selected workplace (BDE)
- List of resources logged on to the currently selected workplace (WRM)
- List of materials/input batches (MPL/TRT) logged on to the currently selected workplace
- List of output batches produced in the currently selected operation (MPL/TRT)

In case you have enabled several lists, you can switch between these lists in the header line that is located above the third list. Activated lists can be selected one after the other.

Maintenance status

If you have purchased the license for the maintenance calendar, the maintenance status is displayed using a yellow or a red field showing a wrench. The color displayed depends on the required maintenance activity.

Calling functions

The functions available are assigned to the relevant objects. Example: The functions *Log person off* and *Log all staff off* are displayed if you click on a person logged on.

3.3 Icon view of workplaces

You can enable this view in the configuration of terminals via the client. Then open this view by clicking the button "< Overview" in the main view. This view shows workplaces in a clear structure and with an image. It shows important information on the single workstations:

Workplace/machine 33021 Status PRODUCTION Yield 833 Scrap 4	Workplace/machine 40312 Status OUT OF MATERIAL Yield 1381 Scrap 5	Workplace/machine 406457A Status PRODUCTION Yield 2500 Scrap 13	Workplace/machine 406457B Status PRODUCTION Yield 66 Scrap 0	Workplace/machine 79667 Status PRODUCTION Yield 51 Scrap 1	Workplace/machine 79668 Status PRODUCTION Yield 78 Scrap 0
--	--	--	---	---	---

Please note: The actual display can be different to the above illustration.

A colored bar to the left of the image indicates the current status of workplaces/machines:

- green: production
- yellow: assigned status
- red: not assigned

Each tile includes the workplace/machine number, the status (text) of the workplace, the yield and scrap quantity and the image of the workplace.

A colored background with a caliper and/or wrench to the right of the image indicates if an inspection or maintenance is due for the workstation.



If you enable the extension [aipkpi](#), the application also shows the KPIs OEE, utilization efficiency and scrap ratio.

The application calculates the KPIs at cyclic intervals (scheduler job "MDE keyfigure calculation"). The KPIs always refer to the current shift. The application calculates the KPIs based on the formulas that are also used for the OEE report and/or the efficiency report on the MOC. The application shows the KPIs with two decimal places. To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached.



The AIP calculates and updates data at cyclic intervals. This may result in deviations between the collected values and the displayed KPIs.

If you click on a tile, the previously described main view is displayed and the workplace is automatically selected. From there, you can perform the postings for the selected workplace.

Use the option "< Overview" to exit the main view and to return to the icon view.



As part of the advanced configuration options, you can customize the layout of display lists, the displayed data fields and functions. For technical reasons, however, you cannot change the sort sequence of display lists in the main view of the terminal.

As of AIP 8.2.2.28, you can automatically change from the main view with tiles to the icon view of workplaces after a configured time. The configuration `AUTOMATIC-CHANGE-TO-START-DISPLAY` is described in the document [AIP2_Configuration_hytnrcfg.pdf](#).

4 Basic Screen as List View

The new tile design can be disabled for the AIP2 terminal. This chapter describes the basic screen with disabled tiles.

In general, the AIP2 has been designed for entries to be made via touch screen. The corresponding functions can be started, selected or executed by touching the buttons or using the displayed virtual keyboard. Selection lists are provided in many cases, as an alternative to manual entries. Required entries can easily be selected from these lists.

Barcodes can be imported/entered in the current dialog using barcode readers, handheld scanners, or swipe card readers. Subject to the barcode prefix, certain data (e.g. operation data) can directly be assigned to the corresponding input field, without having to focus this input field explicitly.

It goes without saying that mouse and keyboard may also be used.



To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

4.1 Basic screens – header and footer

Header



The AIP logo is displayed top left of the screen, which may be replaced with a customer logo after corresponding configuration.

Possible messages are displayed to the right of it (e.g. if a dialog is opened for more than five minutes).

A separate window opens to display error messages that occur during data collection (e.g. validity checks).

Basic screens

A maximum of 16 workplaces or machines can be assigned to the AIP2 terminal. The single workplaces can be found within the list area in the order assigned to the terminal via the client. .

As regards the basic screen of the AIP2 terminal, the user can choose between a tabular view, field-related view and an icon view. This can be configured via the configuration of terminals in the client. The single basic screens are described in the sections that follow.

Footer



The MPDV logo can be found at the bottom left of the AIP2 terminal. Double clicking the logo opens the info dialog where further administration functions can be started. This dialog closes automatically after approx. 5 seconds.

Further information is displayed in the center : the current terminal status, AIP2 version number, date of the build, IP address of the server as well as the terminal number.

The current date and time are displayed to the right.

Terminal status

	Network connection has been established The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.
	The terminal is sending data to the server.
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information, are disabled. But certain postings can be recorded anyway. These postings are transferred to the server, once data connection has been established.
	Data are being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending stored data records to the server.
	DEMO mode The terminal is in the DEMO mode, i.e. server communication is disabled.

4.2 Basic screen “tabular view“

1st list
Workplaces assigned to the terminal

2nd list
List of registered operations

3rd list (optional)
e.g. list of registered staff

The screenshot shows the AIP (Advanced Information Processing) interface. It contains three main sections:

- Machines/workplaces:** A table with columns: Machine/Workplace, Status, Status since, and Note. It lists four entries: 33021 (Manual Workplace Deburrin...), 40312 (Manual Workplace Deburrin...), 406457A (Mounting Place 1), and 406457B (Mounting Place 2). The status is highlighted in green for 'PRODUCTION' and yellow for 'OUT OF MATERIAL'.
- Operations on workplace:** A table with columns: Article, Order, Operation, Target qty., Yield, Scrap, N, and T. It lists one entry: SAR-VPO011-M-9005, S3000209, 0200 Deburring, 1200 PCS, 660, 0.
- Registered persons:** A table with columns: Person, Name, First name, and Advance logon. It lists four entries: 10002630 (White, Susan), 85741 (Meyers, Joseph), 40563 (Green, Mary), and 10002854 (Baker, Jake).

At the bottom, there are buttons for 'Staff work time logging' and 'Staff logging', and a date/time stamp: 23.11.15 16:15:15.

Subject to the configurations made, the tabular basic screen consists of two or three tables. While the first two tables are always displayed, it is up to the user whether or not the third table is shown (optional).

“Machines/workplaces” table

The upper table shows the workplaces assigned to the terminal. The following columns are displayed.

Machine/workplace

The machine or workplace number as well as a description are displayed.

Status

The status is highlighted in color and the status text is shown. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

Status since

Point in time since the status is available.

For ADE workplaces² the point in time refers to the last manual status change. For MDE workplaces it refers to the time when the last status change was identified (for machine connections). It also refers to the point in time when the status was changed manually most recently or to the time of the last shift change.

Please note:

It is indicated here if the “lock production status” function is enabled for the machine/workplace.

Below the first list there is a row including the function buttons mainly relating to machines/workplaces. These functions are described in more detail in the sections that follow.



By way of “customizing” services it is possible to adapt the layout of the display lists, displayed data fields, sort sequences, etc. according to the customer’s requirements. For technical reasons, however, the sort sequence of display lists may not be changed in the basic screen of terminals. The software does not allow it.

Provided that the “compensate manual quantities” option (e.g. set off scrap against yield) is enabled and the machine list also shows shift-related quantities (no default setting), they will not be updated immediately. Quantities are only updated once the lists have been reloaded.

"Operations at workplace" table

The second table shows the operations currently logged on to the selected workplace. The following columns are displayed:

Article

Article defined for the operation

Order and operation

Order number and operation number of the registered operation. Together they build the *MES order number*.

Target quantity

Target quantity defined for the operation.

² We talk of an MDE workplace if this workplace is assigned to a terminal, which runs in the “MDE” operation mode. In any other case, it is an BDE workplace.

Yield

Yield already produced for this operation. The counters of possible machine connections are considered as well.

Scrap

Scrap quantity already produced for this operation. The counters of possible machine connections are taken into account as well.

N

It is indicated here if a note visible on the terminal has been recorded for this operation in the graphic planning board of the client. The note(s) is/are displayed by clicking the OP info button ().

T

If a long text is defined for this operation it is indicated here. The long text is displayed using the OP info dialog (button ).

Below the second list there is a row that mainly includes function buttons relating to operations.

"3rd list" table

The third list is optional and may be configured. Information displayed in this list depends on the workplace configuration.

The following lists can be displayed:

- List of staff logged on to the currently selected workplace (BDE)
- List of resources logged on to the currently selected workplace (WRM)
- Materials/input batches logged on to the currently selected workplace (MPL/TRT)
- List of output batches produced in the currently selected operation (MPL/TRT)

The buttons below the third list (to the left) allow switching between these lists.

Please note

The staff logged on displayed in the third list is identical to the list displayed in the dialog "F5 staff logged on...". Selecting a person in the third list does *not* affect the selection of the operation in the list of OPs running at the workplace. Therefore, it neither affects pre-assignment of the operation in the corresponding posting dialogs.

Toolbar in the basic screen

A toolbar, which may be customized, is assigned to each list included the basic screen. This makes the purpose of a function clear to the user. The “partial upload/confirmation” function can be found below the list of registered operations.

In fact, the toolbar may include several “tabs”, which can be made visible by scrolling to the right/left at the right/left end of the toolbar. A posting dialog (e.g. *change partitioning*) can be opened by clicking the corresponding button.

Please note

The displayed buttons depend on the context defined by the respectively selected workplace. Thus, the displayed buttons may vary when selecting another workplace/machine.

4.3 Basic screen "machine overview"

If the “change view” button is clicked in the basic screen, the view changes to the following presentation:

Toolbar of the assigned machines

Workplace/Machine	33021	Status	PRODUCTION
Short description	WPL02	Start	06:00 18.08.2015
Group	FINISH	Duration [hh:min]	2338:16
Part.	1	Yield	833 PCS
Target cycle [sec/cycle]	20	Scrap	4 PCS
Actual cycle [sec/clock]	0	Cycles	
		Note	

Machine information

Operation	S30002090200	Target qty.	1200 PCS
Article	SAR-VPO011-M-9005	Yield	660 PCS
Article Designation	Mirror housing right,	Scrap	0 PCS
Comment 1			PCS
Comment 2		Yield since logon	0 PCS
Planned duration	06:40	Completion	55%

Order information

Buttons: Modify status, Log on operation, Partial confirmation, Interrupt operation, Log off operation

Footer: AIP logo, US flag, 23.11.15 16:16:51

This presentation gives detailed information on a single machine, whereas the above toolbar still provides an overview of all assigned machines and workplaces.

The presentation consists of three sections:

Toolbar of the assigned machines

The color indicates the current status of all assigned workplaces. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

It is possible to switch between machines by pressing a machine icon (requires a touch screen). Using the keyboard, the active machine can be selected by the arrow keys. To do this, the toolbar must be active.

Workplace/machine information

This display area shows information relating to workplaces/machines and shifts about the currently selected workplace.

Order information

This display area shows information on the registered order/OP. If several orders/OPs are logged on to the workplace, then extra arrow buttons are displayed. It is possible to switch between individual orders/OPs using these arrow buttons.

Notes on selected fields of the machine overview

Unit for yield and scrap

Provided that no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on the primary quantity unit of the operation is displayed.

Partitioning

The displayed partitioning is calculated as follows:

$$\text{Partitioning} = \frac{TLG_M}{DIV_M} * \left[\frac{TLG_{OP1}}{DIV_{OP1}} + \frac{TLG_{OP2}}{DIV_{OP2}} + \dots \right]$$

TLG _M	Partitioning of the machine (TLG in mnr.lst)
DIV _M	Pulse factor of the machine (IMPFAKT in mnr.lst)
TLG _{OPi}	Partitioning of the individual operation (TLG in anr.lst)

The resulting partitioning is displayed without decimal places, provided it is an integer value. Otherwise, 3 decimal places are shown.

In case partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Having logged off all OPs, the machine continues working with the partitioning of the machine.

Target cycle

The largest target cycle of all operations running at the machine is always displayed in the machine overview of the terminal. If this OP is logged off the largest target cycle of the remaining OPs will be displayed.

In case no OP is logged on, the target cycle from the machine list is displayed. Thus, even after a restart, the terminal can get the target cycle that applied at last.

The largest target cycle is also transferred to MDE for monitoring.

Comment 1, comment 2

These two fields show the user fields 53 and 54 (alphanumeric with 20 characters) of the operation. To be able to edit these fields, a corresponding user field key containing these two fields must be defined for the operation.

Target since logon

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time while the production status was not locked for the machine). No value can be calculated, in case the terminal program has been restarted since the OP was logged on.

Calculation:

$\text{TargetSinceLogon} = \text{NetRunningTime}[\text{sec}] * \text{Partitioning} / \text{TargetCycle}[\text{sec/stroke}]$

NetRunningTime: Time since logon while the production lock has not been set. This calculation does not take into account any breaks defined in the shift model or status times posted on RPA 12 (resource performance account).

Deviation [%]

Deviation (in percent) between the expected target quantity since logon and the quantity which has actually been produced "since logon".

Calculation: $\text{Deviation}[\%] = 100\% * (\text{YieldSinceLogon} - \text{TargetSinceLogon}) / \text{TargetSinceLogon}$

Completion

The bar represents the proportion of "yield", which has been produced until now, compared to the "target quantity".

Machine icon:

Provided that the WRM-WTK license has been purchased, the machine icon may be replaced with a picture showing a yellow or red oilcan. It all depends on the required maintenance activity:



4.4 “Machines as icons” basic display

This view can be configured as the default display in the configuration of terminals at the client. It has the advantage that the user can tell from a distance whether or not all machines are in the “Production” status.

All MDE machines have their own buttons colored according to the corresponding status:

- green: Production
- yellow: Assigned status
- red: Not assigned

The button includes details on the workplace/machine number, the registered operation, the yield and scrap quantities as well as the status (text) of the workplace/machine.

If a button is touched, the “machine overview” basic display is shown for this workplace. From there, postings for the selected workplace can be performed using the standard buttons.

By clicking the “symbol” button (if configured) the view changes from the “machine overview” to the “icon view of machines”.

5 Batch-related Entry Function

5.1 Basic screen

Basic terminal screen when a machine is assigned in batch mode:

The screenshot displays the AIP terminal interface. On the left, a list of machines is shown with their status and counts. The main area is divided into sections for machine details, operations, and input materials.

Workplace/machine			
Workplace/machine	Short name	Group	Status
R000	Roll stand	CBM-1	PRODUCTION
Cycles		Start	Duration [hrs:min]
483		13.08.2015	14:00 3618:32
Target cycle	Actual cycle	Yield	Scrap
0,000	0,000	465	18

Operations			
MES order number	MES order number		
M-RF00010010	MPL000010010		
Aluminium Coil produce	Aluminium Coil produce		
Article	Article		
5468-41-0010	5468-41-0010		
Quantities (target/yield/scrap)	Quantities (target/yield/scrap)		
1000 / 3525 / 18	1000 / 503 / 34142		

< Produced output batche	Staff logged on	Input materials >
Person 00040256	Person 00050201	Person 00085741
Name Pernikova, Lisa	Name Meyers, Harald	Name Meyers, Joseph

The basic display shows the third list "Input materials" for machines for which the "batch management" option is configured. All active input batches of the selected machine are displayed in this list.

5.2 Order postings for operations subject to batch management

OP login with input batches

By clicking the "log OP on" button a workflow including two tabs is opened. The first tab "Select OP" the OP to be logged is used.

Logon operation at workplace R000

1 Choose operation | 2 Register OP

Article	Order	Operation	Target qty.	Yield	N	T
801001.5	10000001	0005 Diffusor Pipe Assembly	1000 PCS	11 PCS	17%	
800001.4	10000002	0004 Assembly	2000 PCS	517 PCS		
802001.4	10000004	0004 Assembly	20000 PCS	0 PCS		
802001.6	10000004	0006 Assembly	2000 PCS	0 PCS		

OP-Number: 100000010005

Cancel [Info] OP Info [Go on]

meciv 11.03.16 11:56:43

The “Log on operation” tab is reached by clicking “Next” where in addition to the selected OP, the defined material components are displayed in a list.



Logon operation at workplace R000

1 Choose operation
2 Register OP

Workplace

R000

Operation

100000010005

Material

801001-5

No. of batches

Status

1



Input batch

Input width

Output width

POS	Input mat. no.	Input mat. des.	Custom. batch no.	Remain...	Unit
1	3846-98	Aluminum roll 800 mm x 1000m x 1 mm	384698003	1000	M
0010	5468-41-0010	Aluminum roll 200 mm x 1000 m x 0,5 mm			

Staff Number

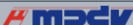
Cancel

Cutting plan

Post batch

Back

Register OP




11.03.16 11:58:24

By entering a batch number in the "Input batch" field and clicking the "Report batch" function, a batch may be logged on as input material for a component. During the entry process, the terminal checks whether the batch number is known to the system and may be logged on. A detailed description can be found in the chapter "Change input batch".

"Batch" field

When an OP is logged on, a batch number is created simultaneously for the next output batch to be produced. The batch number can automatically or manually set (refer to "Settings" in workplace-/resource configuration -> Tab MPL). The batch generated in this way is created with the batch number in the system and set to "running".

If all required input materials are logged on, the OP may be started via the "OK" button in the "OP logon" dialog. Whether input material must be logged on or not, can be defined in the assigned material type of the component.

The screenshot displays the AIP software interface. On the left, there is a navigation menu with options like Overview, R000, R001, P000, 50550, W2030, and F001. The main area is divided into several sections: 'Workplace/machine' showing details for R000 (Roll stand, Status: PRODUCTION, Cycles: 483, Target cycle: 0,000, Actual cycle: 0,000, Yield: 1465, Scrap: 18), 'Operations' showing MES order numbers and articles for Aluminium Coil produce, and 'Produced output batches' showing personnel logged on (Pernikova, Lisa; Meyers, Harald; Meyers, Joseph). The bottom status bar shows the date and time: 11.03.16 11:59:03.

Once the OP has been logged on successfully, all active input batches of the selected machine are displayed in the material list.

Logoff/interruption of OPs

A running OP may be interrupted or logged off by clicking the "logoff/interrupt OP" button. Then a dialog opens, where the following selection can be made:

Interrupt/log operation off
<A_AUT_MPL>

Workplace W2030

Operation M30002000010

Article	Order	Operation	Target qty.	Yield	Scrap	N
4529-42	M3000200	0010 create Solution	130 L	0	0	

Cancel
 Unregister OP Interrupt OP

11.03.16 12:00:53

Using the button "Log OP off" the system opens a screen to do so which includes the same input field as the "Output batch change" (see next chapter).

Thus, the output batch currently active is completed, when OPs are interrupted or logged off.

Interrupt operation
<A_UN_MPL>

Workplace 00050000

Operation 500000000100

Current batch PR1MF6S111111112

Material 611-2

Target

Transp. unit

Comment

Quality ☒ Yield ☐ Scrap

Quantity PCS

Reason

Interrupt. reason

Deviation reason

Staff badge number ☒ Memorize

Cancel
 Preced. batches Change i.batch Interrupt OP

5.3 Batch-related postings

Input batch change

Using the button "Change input batch" a change of input material can be changed during a running OP.

<CE_WL_MPL>

Workplace R000

Operation

Material 5468-41-0010

Input batch

POS	Input mat. no.	Input mat. des.	Custom. batch no.	Remain...	Unit
1	3846-98	Aluminum roll 800 mm x 1000m x 1 mm	384698003	1000	M
0010	5468-41-0010	Aluminum roll 200 mm x 1000 m x 0,5 mm			

Staff Number

Cancel
✓ Post batch

11.03.16 12:04:49

Log off input batch

Input batches may be changed by entering a currently active batch number or by entering a new batch number. When logging batches off, it is also possible to enter the status and consumption of the batch to be logged off.


Unregister Incoming Batch
<CE_AB>

Batch	384698003		
Material	3846-98		
BOM item	1		

Status
☐ PROCESSED
☐ LOCKED
☒ with remaining qty.

Consumption 10 M

Comment

Staff Number


Unregister Incoming Batch
03.07.14 13:03:20

Options when logging input batches off:

F1 - PROCESSED

The batch is set to status "Processed" and the remaining quantity is set to 0. A consumption posting as goods issued is generated for the remaining quantity.

F2 - BLOCKED

The batch is set to the "blocked" status. A consumption entered additionally is deducted from the current, remaining quantity as goods issue.

F3 - with remaining quantity

The batch is set to the "free" status. A consumption entered additionally is deducted from the current, remaining quantity as goods issue. If the remaining quantity that is still available becomes ≤ 0 , the batch status automatically switches to "processed".

Consumption

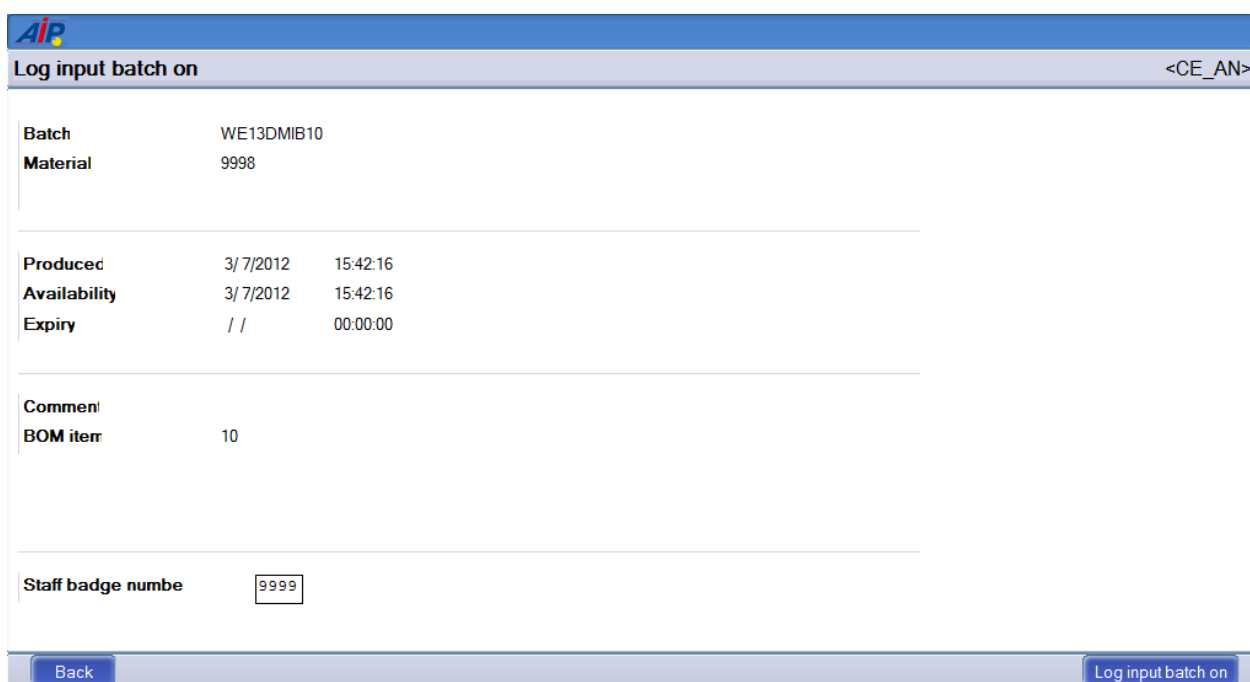
The entered consumption (unit of the input material) is deducted from the remainder of the batch and a goods movement is generated.

Comment on batch

The comment entered is saved as information for the batch.

Log input batch on

Provided that the batch is known, batch data is displayed in an intermediate dialog where the logon may be confirmed.



Batch		WE13DMIB10	
Material		9998	
Produced	3/ 7/2012	15:42:16	
Availability	3/ 7/2012	15:42:16	
Expiry	/ /	00:00:00	
Comment		10	
BOM item		10	
Staff badge number		9999	

If the batch could be logged on, it is taken over to the material list in "customer batch number" and thus the change is completed.

However, in case the logon is inadmissible as the input material does not correspond to that of the component, the logon is rejected by the following an error message.

Logon of unplanned input material

In addition to planned materials, it is also possible to log on "unplanned" material for an OP, using an additional feature for the OP. If the "replaceable" option is set to "J", the user is able to assign the respective component manually, when batches are logged on. However, the logon is only allowed if the material type of the input batch matches that of the component.

In the selection list the components are filtered by the material number and displayed in green for selection.

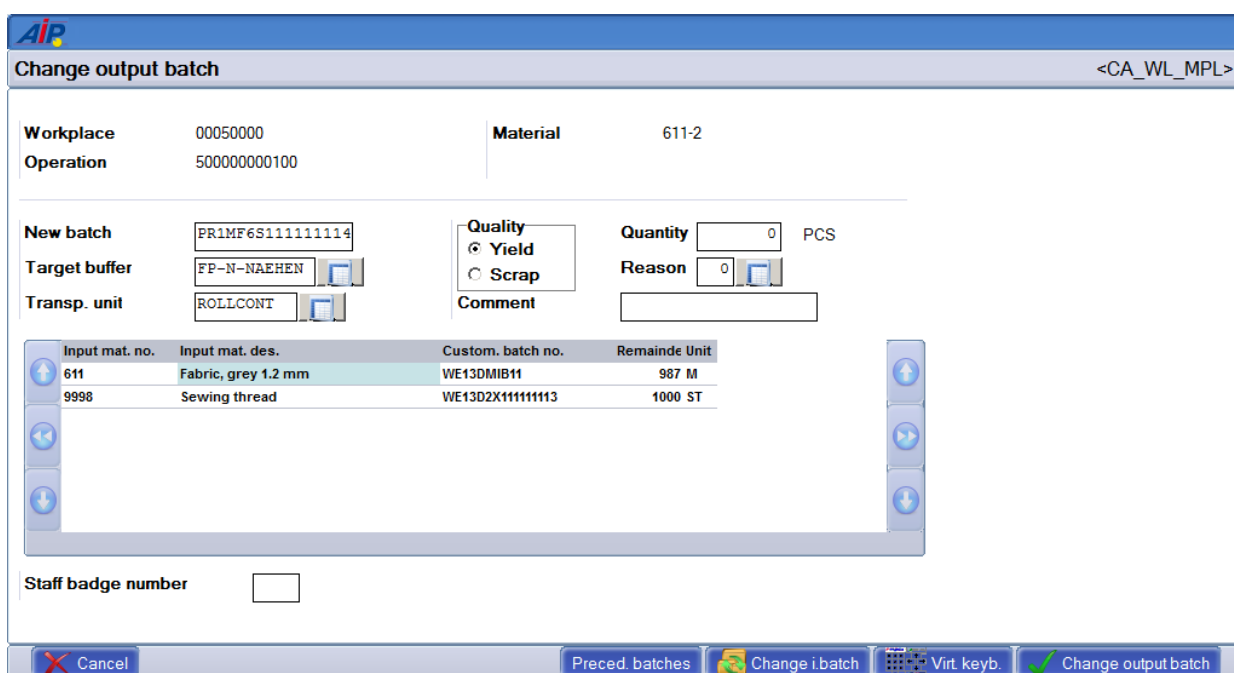
Logon of unknown batches

Using the option "Create unknown charges" in the HYDRA basic settings (Tab BDE-> Settings 2) it is possible to register an input batch still unknown to HYDRA to an OP.

In this case, when input batches are logged on, the system searches for a valid assignment of input material to the material type of the selected component. If a matching assignment is found and the "allow entry of unknown input batches" option is configured for the material type in the "input batch processing" tab, the batch is generated by logging it on to the system. Then it is set to the "running" status. The batch is initially created in a quantity of 1.000.000.000.

Output batch change

Output material may be changed for a running OP using the "output batch change" option.



Change output batch <CA_WL_MPL>

Workplace: 00050000
Operation: 500000000100
Material: 611-2

New batch: PR1MF6S111111114
Target buffer: FP-N-NAEHE
Transp. unit: ROLLCONT

Quality: ☒ Yield ☐ Scrap
Quantity: 0 PCS
Reason: 0
Comment:

Input mat. no.	Input mat. des.	Custom. batch no.	Remainde Unit
611	Fabric, grey 1.2 mm	WE13DMIB11	987 M
9998	Sewing thread	WE13D2X111111113	1000 ST

Staff badge number:

Buttons: Cancel, Preced. batches, Change i.batch, Virt keyb., Change output batch

The input batches that are logged on are displayed with their available remaining quantity in a list within the output batch change dialog. The following data may be entered:

Target buffer

Material buffer for which the current batch is to be produced. The output material buffer of the machine is pre-assigned as default value.

Transport unit

A transport unit defined within the system may be assigned here. The selection refers to transport units that were assigned to the material type of the OP.

Comment on batch

In this field, a comment may be saved for the produced batch.

Quantity

The batch to be produced is posted with the quantity entered here. The quantity is taken over as primary quantity of entry to the order and machine and a goods movement is generated as goods receipt.

Quality

A batch may be classified as yield or scrap quantity. The system posts yield batches with the "free" status. Scrap batches automatically get the "blocked" status. When scrap is selected, a valid scrap reason has to be assigned.

"Preceding batches" function key:

This button opens a list with output batches which have already been produced for this OP.

"Change inp. batch" function key:

Using this button the user can switch to the "input batch change" function.

New batch

When a current output batch is completed, a new batch number is simultaneously created for the next batch. The batch number may be assigned automatically or manually. The batch generated in this way is created with the batch number in the system and set to "running".

Entry of additional batch attributes

Several additional batch attributes may be recorded for a material type by configuring MPL --> Master data --> Attributes.

In case attributes are defined to be recorded at the terminal for the material type of the running OP, another input dialog is opened when the output batch is changed. The screen opens additionally after clicking OK in the output batch change, interrupt OP and finish OP function.

Numeric and alphanumeric values, which are then saved for the produced batch in an additional table, may be recorded via batch attributes.

Recording Goods Receipt Batch

A new goods receipt batch may be created in the system via the "enter GR batch" button.

Register Goods Receipt Batch
<C_GEN>

Batch
WE1WEE2R112

Workplace
R000

Operation
M30002000010

Material
4529-42

Target buffer
MP_V_50506

Transp. unit
CT_15

Comment
Supplier batch

Quality
Yield
Scrap

Qty
100
PCS

Reason
0

Staff Number
9999

Cancel
Enter batch GR

Having clicked the "OK" button, the batch is created and the dialog remains open for further entries.

Batch numbers may be generated automatically or manually depending on the configuration. In addition, the following data is saved at the batch.

Workplace

Machine where the batch was recorded

Operation

Order where the batch was recorded

Material

Material number of the batch

Quantity and unit

The batch is created with the quantity and unit entered here. A goods receipt is posted with the quantity.

Quality

A batch may be classified as yield or scrap quantity. The system creates yield batches with the "free" status. Scrap batches automatically get the "blocked" status and the batch class "scrap". When scrap is selected, a valid scrap reason has to be assigned additionally with respect to the corresponding workplace.

Target buffer

Material buffer for which the current batch is to be produced. The output material buffer of the machine is pre-assigned as default value.

Transport unit

A transport unit defined within the system may be assigned here.

Comment on batch

In this field a comment may be saved for the goods receipt batch.

Repost Batch

Using the "repost batch" button, an existing batch may be reposted to another material buffer.

Re-post Batch

<C_UMB>

Batch			
Workplace	R000		
Operation	M-RF00010010	⏏	
Material	5468-41-0010		

Target buffer	MP_N_50505	⏏	
Comment			

Quality	Qty	0	
☒ Yield	Reason	0	⏏
☐ Scrap			

Staff Number	9999
--------------	-----------------------------------

Having clicked the "OK" button, the batch is reposted to a new material buffer and the dialog remains open for further entries.

As an alternative, the batch can also be reposted from yield to scrap.

5.4 Run-through batch mode

A HYDRA machine may also be configured in "run-through batch mode". In run-through batch mode input material is continued being processed with unchanged number (run-through batch number) via an OP.

The entry functions for throughput batch processing at the terminal are identical to those for active batch tracing at the machine.

Please note: At machines with "throughput batch mode", it is impossible to log operations on at the same time.

5.5 Manual report batch quantity in the MPL environment

Using the manual partial upload function of HYDA-ADE (A_TR dialog), it is possible to record a partial quantity for the current operation and thus to the active output batch.

The following performance as regards the entry of quantities results from a manual partial upload/confirmation:

- Scrap is only transferred to the active operation and not to the output batch.
- Yield is booked cumulated onto the current output batch.
- In this case, log records of the record type "H" are also generated when the shift changes, provided that yield has been recorded before
- In case yield has already been recorded as partial quantity for an output batch, it is no longer possible to log the batch off as scrap when the output batch is changed the next time. In this case, the batch assigned to the "yield" class has first to be completed, before scrap can be posted again.

Effects on available retrograde material components are as follows:

- When a total quantity is recorded, scrap is not deducted in a retrograde manner

- When yield/scrap is recorded, it is also withdrawn when it comes to scrap
- Negative values, which might result from offsetting quantities, are not considered

The following notes and restrictions are to be taken into account:

- Personal partial uploads (with P_AB) are not taken into account
- Scrap quantities are only uploaded to a PPS system, provided that the interface is set to "upload of partial confirmations"
- Now only quantities of the accounts yield and scrap are supported.
- The partial upload itself does not trigger a goods movement. This is only the case when the output batch has been completed.
- The checks made with respect to the input quantity and display within the consumption balance cannot be used together with the "collect input quantity in relation to batches" configuration within the material type, as the current output batch is not completed for partial quantities. This affects also the collection of serial numbers together with batches (MPL-SNR).

5.6 Display produced output batches

It is possible to display the output batches produced for a running operation that is subject to batch management using the [3rd list of the machine master](#).

A default number of 20 output batches is displayed for each machine in the list. Output batches (yield and scrap batches) are shown, which have been produced at this terminal since output batches were changed.

As the terminal only keeps this list locally, it is not synchronized with the server, when AIP is started.

The list includes, among other things, the article, article designation, batch number, date, time, quantity and batch class, user fields as well as alternative batch numbers. Some of the fields might not be completed depending on the data collection scenario when generating output batches.

List contents can be configured via the [MNR_AMAT.LST] section in the ctaipplay.ini file.

Configuration:

List contents can be configured via the [MNR_AMAT.LST] section in the ctaipplay.ini file.

Example:

CTAIPLAY.INI
<pre>[MNR_AMAT.LST] GRID_FONT=Arial GRID_FONTSIZE=9 GRID_COLOR=clBlack GRID_BACKGROUND=clWhite GRID_ORDER=ROW.IDX=- GRID_CAPTION=Output batches produced EXAMINE_SCANEXPR1=KLASSE=G EXAMINE_SCANCOLOR1=clGreen EXAMINE_SCANEXPR2=KLASSE=A EXAMINE_SCANCOLOR2=clRed ; ROW.IDX=N10,50,R,Row CNR=C20,150,L,Losnummer ;CNR=*CNR,Batch ATK=C25,125,L,Artikel ; ATK=*ATK ; KLASSE=C3,40,Z,* MENGE=N12.0,70,R,Menge EINH=C3,30,Z,ME DAT=dd.mm.yyyy,70,L,Date ZEI=hh:mm:ss,60,L,Time ATKBEZ=C30,200,L,Artikelbezeichnung</pre>

Display as third list:

< Input materials		Produced output batches			Staff logged on >
...	Batch PR4RD97117	Batch PR4RD97116	Batch PR4RD97115	Batch PR4RD97114	>
	Article 5468-41-0010	Article 5468-41-0010	Article 5468-41-0010	Article 5468-41-0010	
	Aluminum roll 200 mm x 1000 m	Aluminum roll 200 mm x 1000 m	Aluminum roll 200 mm x 1000 m	Aluminum roll	

Figure: Display of the output batches produced

Using the caption of the 3. list, the user can switch to the display of the produced output batches.

Server based comparison of produced output batches

From AIP program version V#2.0.54 the system can synchronize the list of the produced output batches of a machine with the server using a cyclic comparison.

The server-based comparison is activated [\[Technical: PDM list 13 with MOD=P\]](#) by an entry [Configuration](#) in the customer-specific configuration file „ctlisten.ini“.

The system only carries out comparisons done for output batches collected over the last 3 days in this terminal.

5.7 Batch information

Information on a batch may be displayed in a dialog via the "batch info" button.

Batch information		<C_LOS_INFO>
Batch	PR4RD96114	
Material	5468-41-0010	
Designation	Aluminum roll 200 mm x 1000 m x 0,5 mm	
Type	SYSTEM	
Class	G	
Operation	M-RF00010010	
Machine	R000	
Material Buffer	MP_N_50500	
Balance	0,0	
Prod.date	12.10.2010 15:45:27	
Status	F	
Note		

OK

5.8 Display of consumption balance

When logging an OP off, the “consumption balance” (V_B LZ dialog) can be displayed, which can be configured via the machine and the material type of the operation.

The consumption balance is shown if this option is active at the machine and material type.

The consumption balance shows the material consumption based on batches and the user is able to log still running batches off.

Configuration:

A special configuration (function = “DLG=V_B LZ;BREAK-ON-CANCEL“) has to be defined for the OK button of the logoff dialog (e.g. A_AB_MPL) to make sure that the consumption balance is started when the OP is logged off.

Sample configuration of the OK button for starting the consumption balance:

Display of consumption postings

The “show details” function allows for consumption quantities, which have been collected so far, to be displayed.

Logging input batches off

By way of the “log batch off” function, the user can choose a currently running batch from the list and log it off by entering a consumption.

5.9 Advance logon of input batches

The AIP supplies the following dialogs of functions to issue an advanced logon: The function for logging input batches on in advance is used to set up and log on the next input batch while an OP and input batch are still running. This next input batch is not yet running but assigns the “logged on in advance” flag.

An input batch may be logged on in advance for a currently running OP or a prepared OP.

General / usage

The process might require an input batch to be logged on in advance and set up accordingly on a machine, while the preceding input batch is still being used for a material.

This situation frequently occurs at very large machines processing, for example, roles or belts that are uncoiled as input batch at the beginning of the machine and coiled up as output batch at the end of the machine.

As the users are mostly busy with activities at the end of the machine at the time when the input batch actually needs to be changed, they cannot perform the input batch change and, as a result, they are provided with the opportunity to log the next input batch on already in advance for an order/OP.

Then the input batch can actually be changed by logging a new OP on or a project-specific call can be established.

Configuration

The settings required for using the function “advance logon of input batches” is described [here](#).

Procedure

The procedure for using the function “advance logon of input batches” or the logical process is described [here](#).

Usage at AIP

Basic AIP view

The basic AIP screen shows the function key “Advance logon of input batch” (advanced logon of input batch). The dialog for logging input batches on in advance may be used by clicking this function key.

Advance logon of input batches (CE_VWL_MPL)

The user highlights in the display the workplace where input batch is logged on in advance and pushes in the 3. list the function button with three points. The dialog (CE_VWL_MPL) opens by clicking the function key “Advance logon of input batches”.

If an operation is currently running/logged on the workplace, this one will be selected by default. The input batch (that is to be logged on in advance) is entered/scanned for the selected BOM item. Advance logon of input batches is started by clicking the button “post batch”.

At first, the input batch is checked for validity (dialog CE_VAN). The material number of the input batch is checked against the material number of the component list or the BOM item. The input batch is logged on in advance, once the button “log input batch on in advance” has been clicked:

Preregistration of input batch <CE_VWL_MPL>

Workplace: CRM Material: CCSD1901223

Operation: 22999998010

Input batch:

POS Input mat. no.	Input mat. des.	Cust. Batch nr.	Remainde Unit
01 A06A2100926		R835-022	18 mt

Staff badge number:

Cancel Post batch

CTAIP V2.0.2.67 Server:win2008-10-3 TNR:999 26.03.13 12:22:46

Finally, the input batch that has been logged on in advance is displayed in purple in the BOM of the component.

The dialog is exited by the "cancel" key.

Display of the third list

The system displays as the result the input batch logged on in advance additionally to the logged on/ running input batches in the third list of the basic display with the icon

A BOM item can then simultaneously use a logged/ running input batch and a input batch logged in advance.

Log off an input batch logged in advance

It is still possible to log off or reset an input batch logged in advance in the dialog "Advance logon of input batches".

The user highlight again the input batch logged in advance and clicks on "Report batch".

The input batch is logged off using the dialog CE_VAB.

To log on an "Advance logon input batch" to an OP

The system displays in a BOM item an input batch logged in advance during the logon of OP in production. During OP logon, the input batch logged in advance is automatically logged.

But if another input batch is logged to that BOM item, then the input batch logged in advance remains applicable to the running OP.



During logoff / interruption of an OP, all input batches logged in advance are automatically logged off / reset.

5.10 Collection of serial numbers

Serial numbers are assigned to be able to differentiate between individual items of material. The combination material and serial number identifies a single part.

MES provides several variants with different features to record serial numbers:

- Variant 1: Entry of serial numbers for OPs that are not subject to management in batches (dialog A_SNR/ "E" only)
- Variant 2: Entry of serial numbers for OPs that are subject to management in batches (dialog A_SNR)
 - Manual input of the serial number. The serial number is assigned to a HYDRA batch number.
 - Automatic assignment of the serial number = HYDRA batch number
 - Automatic assignment of the serial number. The serial number is assigned to a HYDRA batch number.

5.10.1 Entry of serial numbers for OPs that are not subject to management in batches (dialog A_SNR/ "E")

General / usage

Serial numbers are assigned in ERP when creating the production order. At the interface to HYDRA, assigned serial numbers are transferred as details for the order header and managed there.

The OP that is not subject to management in batches of the order is logged on and respective serial numbers are recorded in the relevant quality (yield/scrap) at AIP.

The serial number is entered manually (e.g. also by barcode on the single part) by the AIP user. To facilitate data collection, the serial numbers available for the order can also be displayed in a list. Only serial numbers may be logged on that have been assigned for this order. Please note that a serial number may exist as scrap only once.

The quantity recorded for the order/OP is always 1 for each single part and, depending on the quality, posted to the order's/OP's yield or scrap account.

Uploads to the ERP system in relation to the serial number (goods movements) can be performed for recorded serial numbers/single parts.

Configuration

The settings required to use the function for collecting serial numbers (with orders subject to management in batches) are described [here](#).

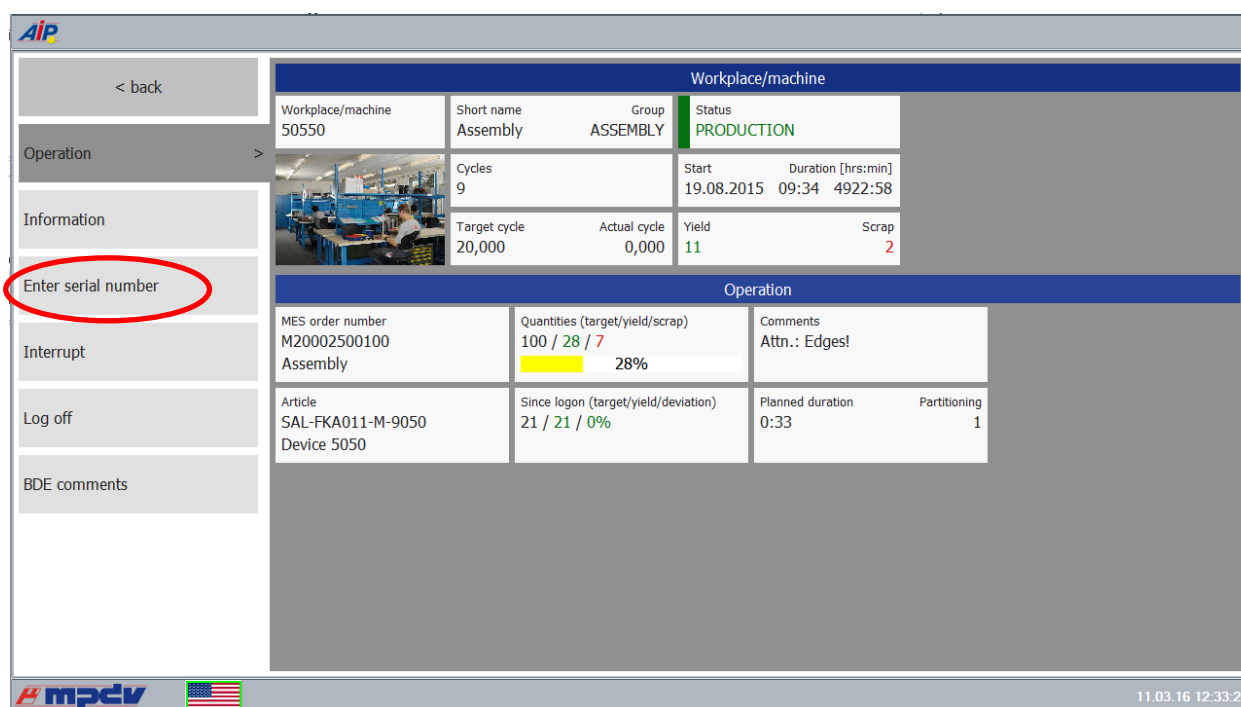
Procedure

The procedure required to use the function for collecting serial numbers (without orders subject to management in batches) or the logic process are described [here](#).

Usage at AIP

Basic AIP view

The user applies the "serial numbers" dialog (A_SNR). The function key is configured for output batches in the layout. The layout opens when pushing the function key with three points in the 3rd list of "produced output batches".



The screenshot displays the AIP interface with a sidebar on the left and a main content area. The sidebar contains the following options: "< back", "Operation", "Information", "Enter serial number" (highlighted with a red circle), "Interrupt", "Log off", and "BDE comments". The main content area is divided into two sections: "Workplace/machine" and "Operation".

Workplace/machine section:

Workplace/machine	Short name	Group	Status
50550	Assembly	ASSEMBLY	PRODUCTION

Operation section:

MES order number	Quantities (target/yield/scrap)	Comments
M20002500100 Assembly	100 / 28 / 7 28%	Attn.: Edges!

Article section:

Article	Since logon (target/yield/deviation)	Planned duration	Partitioning
SAL-FKA011-M-9050 Device 5050	21 / 21 / 0%	0:33	1

The bottom of the interface shows the "mpcdv" logo, a US flag, and the date/time "11.03.16 12:33:26".

Display: Layout to request function key "serial number" (A_SNR)

Collection of serial numbers (A_SNR)

Figure: Collection of serial numbers at the terminal (A_SNR/ "E")

Serial number

The user enters the serial number. The serial number must be part of the order. A validation check is performed. If the serial number is not part of the order or if it is already recorded with a result, it cannot be used.

Clicking the function  opens a selection list showing the "free" serial numbers of the order. Requesting the selection list simultaneously updates the list of already recorded serial numbers.

Grid

The displayed list shows all serial numbers in their relevant quality (yield/scrap) that have already been entered.

- Scrap is displayed in red.
- Yield is displayed in green.

Quality

When a serial number is entered, it has to be assigned a quality. You may choose from the following options:

- Yield
The serial number is entered with "yield" quality. The operation quantity is increased by yield = 1. The default value is "yield".
- Scrap
The serial number is entered with "scrap" quality. The operation quantity is increased by scrap = 1.
- Scrap reason
If the "scrap" quality is entered, the user also has to enter or select a scrap reason.

Entry function

By clicking this function, the entered serial number is recorded with the selected quality.

List function

By clicking this function, the entire list of already entered serial numbers may be updated by the selected quality (e.g. when the dialog is reopened).

Interrupt/terminate OP for orders with serial numbers (A_UN/A_AB)

Quantities are not entered when operations requiring serial numbers are logged off. The quantity fields must be disabled in these dialogs.

5.10.2 Entry of serial numbers for OPs that are subject to management in batches (dialog A_SNR)

General / usage

The following alternatives arise for entering and/or assigning serial numbers if an order/operation subject to batch management is used to enter serial numbers:

- Manual input of the serial number. The serial number is assigned to a HYDRA batch number.
- Automatic assignment of the serial number = HYDRA batch number

- Automatic assignment of the serial number. The serial number is assigned to a HYDRA batch number.

Configuration

The settings required for the function to collect serial numbers (with orders subject to management in batches) are described [here](#).

Procedure

The procedure required in order to use the function for collecting serial numbers (without orders subject to management in batches) or the logic process are described [here](#).

Usage at AIP

Collection of serial numbers (A_SNR) including manual assignment of serial numbers - type "E"

An additional batch is created for each registered serial number in the area of material and production logistics if batch management requirement is enabled for the operation.

The connection between the current output batch and the registered serial number is saved additionally in the database for traceability purposes.. In this case, the output batch is considered as ID without inventory and, therefore, does not receive a quantity.

For OPs handled in batches, batch attributes that might have to be recorded are entered using the general standard dialog prior to sending them.

The serial number can be entered. Batches are assigned in the background.

Collection of serial numbers (A_SNR) including automatic assignment of serial numbers - type "G"

If the option "serial number requirement = G" is set for the operation, output batches will be recorded as serial numbers. In this case, the serial number is the output batch number.

For OPs subject to batch management, the dialog and relevant entries are as follows:

Collection of serial numbers (G)
<A_SNR>

Workplace
Operation

Material

M20002500100

SAL-FKA011-M-9050

Serial No.

Serial number	Class	Date	Time	*	Reason
PR4RH6B11G	A	04.07.2014	08:08:17	2	COLOR DEFECT
PR4RH6B11F	A	04.07.2014	08:08:09	1	BURR FORMATION
PR4RH6B11E	G	04.07.2014	08:07:46	0	
PR4RH6B11D	G	04.07.2014	08:07:45	0	

Quality
☐ Yield
☒ Scrap

Reason

Staff Number

✕ Close
List
Capture

04.07.14 08:09:38

The "serial number" field is disabled and includes the current output batch. Once the badge number has been entered, posting can be performed by clicking "capture". If the entered quality is not "yield", a valid scrap reason has to be entered.

Exactly one batch with quantity 1 is created for each serial number and the serial number is the batch number.

The "list" function updates the list of already recorded serial numbers.

For OPs handled in batches, batch attributes that might have to be recorded are entered using the general standard dialog prior to sending them.

Collection of serial numbers (A_SNR) including automatic assignment of serial numbers using the number range - type "S"

Serial numbers are recorded in relation to the batch if the option "serial number requirement = S" is set for the operation. In this case, the operation must be subject to management in batches. Every registered serial number causes an output batch to be changed (CA_WL) creating a batch with quantity 1 regarding the serial number in the MPL module.

The server determines a new serial number and displays it in the "serial number" field by clicking the



function . The new serial number is assigned uniquely for the whole system using the "SNR" number range.

For OPs handled in batches, batch attributes that might have to be recorded are entered using the general standard dialog prior to sending them.

Interrupt/terminate OP for orders with serial number tracking

Quantities are not entered when operations requiring serial numbers are logged off. The quantity fields must be disabled in these dialogs.

6 Barcode Input with Prefix

A barcode is interpreted as prefix barcode if the third character is a dot. In this case the first two characters identify the barcode type. The actual barcode starts with the fourth character.

ID+ Prefix	Example	Comment --> processing
-----	-----	---- General ---
00.	00.ABC123	Data not defined, → 00. will be deleted and "ABC123" will be passed on to standard processing
01.	01.OK 01.ESC	Action barcode → Dialog cancelled or ended with OK button or Esc button.
-----	-----	---- HYDRA-ADE + HYDRA-LLE + HYDRA-MDE ---
10.		(combined) Order/sequence/OP number → acronym <ANR>
11.		Order (header) → acronym <AUNR>
12.		Sequence → acronym <AFOLG>
13.		OP → acronym <AGNR>
14.		Sub-order number -> Acronym <UAGNR>
22.		Split no. → acronym <SPLNR>
15.		Upload/confirmation number → Acronym <RMNR>
16.	16.EXTRUDER-7 16.200	Machine → Acronym <MNR> → Passed on to dialog with MNR=EXTRUDER-7 or MNR=200
17.	17.1 17.1001	Machine status → Acronym <MST> → Passed on to dialog with MST=1 or MST=1001
18.	18.1 18.1001	Scrap reason → Acronym <EGG:AUS> → Passed on to dialog with EGG:AUS =1 or EGG:AUS=1001
19.	19.1 19.1001	Deviation reason → Acronym < EGG:GUT > → Passed on to dialog with EGG:GUT =1 or EGG:GUT=1001
20.	20.1 20.MF	Operator position → Acronym <BPOS> → Passed on to dialog with BPOS =1 or BPOS = MF
21.		Wage and premium indicators → Acronym <LPKZ>
-----	-----	----HYDRA-WRM + HYDRA-DNC + HYDRA-PDV + HYDRA-MPL ---
40.	40,100 40.MONTAGE	Destination → Acronym <ZLO> → Passed on to dialog with ZLO=100 or ZLO= MONTAGE
41.	41.KARTON 41.KISTE	Transport unit → Acronym <TPE> → Passed on to dialog with TPE = KARTON or TPE = KISTE
42.		Batch number → Acronym <CNR>→→
43.		Throughput batch number → Acronym <DLL>→→
44.		Alternative batch number → Acronym <CNR:ALT1>→→
45.		Alternative batch number → Acronym <CNR:ALT2>
46.		Alternative batch number → Acronym <CNR:ALT3>
47.		Alternative batch number → Acronym <CNR:ALT4>
48.		Alternative batch number → Acronym <CNR:ALT5>
49.		Alternative batch number → Acronym <CNR:ALT6>
-----	-----	---- Mainly for the HYDRA-PZE module ---

ID+ Prefix	Example	Comment --> processing
50.		Badge number → Acronym <KNR>
51.		Personnel number → Acronym <PNR>
52.	52.EDV 52.VERTRIEB	Cost center → Acronym <KST> → Passed on to dialog with KST=EDV or KST=VERTRIEB
53.	53.1 53.1001	Absence reason → Acronym <FGR> → Passed on to dialog with FGR=1 or FGR=1001
-----	-----	---- Customer-specific barcodes ---
90.		
...		

In case barcodes are required, which actually have a dot at the third place (e.g. if the machine/workplace number has a dot as third character), it is possible to define an alternative indicator for barcode prefixes in the HyTnrCfg.ini terminal configuration, e.g.

[Terminal->USR 0]

BarcodePrefixChar=\$



If another prefix is actually required, the respective barcode font in use must be able to represent this prefix.

Examples for barcodes

Barcode printing: Font "Codedreineun" and prefix "."

Prefix	Barcode	Raw data
10.	*10.123456780100* *10._ABCD12340100*	ANR = 123456780100 ANR=_ABCD12340100

Prefix	Barcode	Raw data
11.	*11.12345678*	AUNR = 12345678
12.	*12.01*	AFOLG = 01
13.	*13.0100*	AGNR = 0100
14.	*14.0000*	UAGNR = 0000
15.	*15.123456789012345*	RMNR = 123465789012345
22.	*22.02*	SPLNR = 02

Prefix	Barcode	Raw data
16.	*16.123456*	MNR = 123456
17.	*17.1122*	MST = 1122
18.	*18.1234*	EGG:AUS = 1234
19.	*19.123456789*	EGG:GUT = 132456789
20.	*20.13*	BPOS = 13
21.	*21.1221*	LPKZ = 1221

Prefix	Barcode	Raw data
50.	*50.1337*	KNR = 1337

6.1 Configuration of customized barcode prefixes

Section [barcode]	
BarKenn90=SAPCNR BarKenn91=EGR:GUT	The barcode prefixes 90...99 can be assigned here according to the customer's requirements. This means, if a barcode with the relevant prefix is used, it will be transferred to the dialog along with the assigned ID. Then the barcode has the following structure: <Prefix>.<Net barcode> e.g.: "90.12345" → SAPCNR=12345

Firmly assigned barcode prefixes:

10:ANR 53:FGR

11:AUNR

12:AFOLG

13:AGNR

14:UAGNR

15:RMNR

16:MNR

17:MST

18:AUSGRD

19:AGGGRD

20:BPOS

21:LPKZ

22:SPLNR

40:ZLO

41:TPE

42:CNR

43:DLL

44:CNR:ALT1

45:CNR:ALT2

46:CNR:ALT3

47:CNR:ALT4

48:CNR:ALT5

49:CNR:ALT6

50:KNR

51:PNR

52:KST

7 AIP2 -Local Configuration

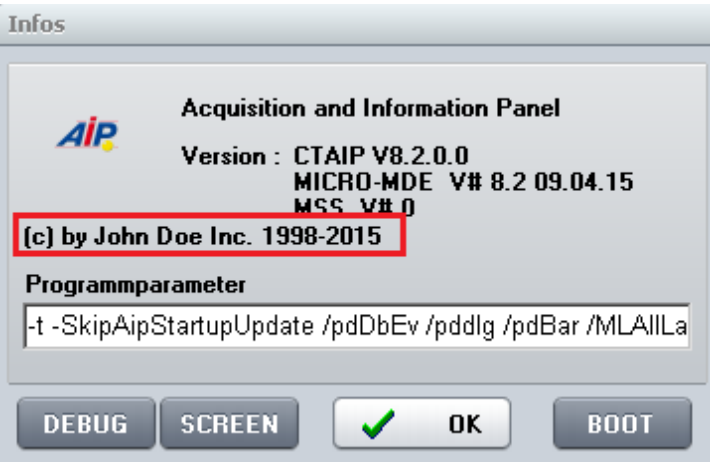
7.1 Local Configuration ctaip.ini

The most important hardware and system settings are defined for each terminal in the CTAIP.INI file of the C:\MPDVAIP2 directory.



Changes to the configuration file ctaip.ini are only enabled after rebooting the terminal software.

Layout configuration

Entry	Comment
Section [system]	
CompanyInfo=(c) by John Doe Inc. 1998-2015	Overwrites the copyright text in the "About" dialog 
parameters= ... -t +t	The virtual keyboard can be disabled/enabled, irrespective of the terminal type.
DEMO_ALL=ON	The information in the lower bar are normally hidden in demo mode. This option, however, causes all pieces of information to be displayed even in demo mode (version number and date, server, terminal number, online lights).
parameters= ... - SkipDynDlgExtraction...	Internal option preventing the extraction of dialog fields/buttons when updating dialog data. (Only applies if WF directory ".\spool\%w\$.\$\$\$\" is available)
parameters= ... - SkipAipStartupUpdate...	Internal option preventing the configuration files (*.ini,*.cfg,*.cfi) of the module DLLs (packets*.dll) and application DLLs (functions*.dll) from being synchronized when starting the terminal.
VirtScreenSize=640	All windows are started with the indicated resolution

Entry	Comment
VirtScreenRatio=16:9	The display ratio remains if the configuration VirtScreenSize is used to reduce the window. The width-to-height (aspect) ratio can be changed using VirtScreenRatio. Consequently, the width-to-height ratio 16:9 can be tested with a 4:3 monitor and vice versa. The value can be configured as "16:9" or as floating point value (e.g. 1,77777). Note: Information screens are specifically scaled if monitor screens in portrait format (e.g. 9:16) are used. This special scaling can be disabled in hytnrcfg.ini: [Tnr Konfiguration 0]->PortraitAlignment=off
Section [SKIN]	Configurations for the terminal's skin (see ctaipplay.ini) The online configuration of the terminal can be reached by the info dialog (click MPDV icon). ALT+F1 opens the control elements for the skin. Please note: Only in the design/GUI of AIP 8.1 and/or the dynamic dialogs
Saturation=0	Skin configuration / Default = 0
Hue=0	Skin configuration / Default = 0
Name=mpdv	Skin to be used / Default = mpdv
Active=false	Skins can only disabled in ctaip.ini! Default = true

Entry	Comment
Section [system]	
Ustr=21	Distinct terminal number
Hostname=192.9.200.24	Internet address of the server
Offlinetimeout=600	The interval after which online access should be attempted the next time. The interval is specified in seconds
Showcursor=on	Show or hide mouse pointer in terminal application: on: mouse pointer active off: mouse pointer inactive
Loadfile=ctnet\win\aip2.txt	Configuration file to download the application from the server. The paths is relative to the installation directory of the system.
Watchdog=on	ON: Watchdog is activated OFF: Watchdog is not activated

Entry	Comment
Demo=off	'on': Offline demo mode; always off in the production environment!
parameters=-t	The -t parameter switches off the virtual keyboard.
TMOUT_C=xxx	Timeout for CONNECT to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_S=xxx	Timeout for SEND to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_R=xxx	Timeout for RECEIVE of the server If not specified, default = 120 seconds
TMOUT_F=xxx	Timeout for FILESERVER operations to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
Section [barcode]	Configuration of customized barcode prefixes.
BarKenn90=MNR4 ... BarKenn99=ANR3	BarKenn90 > defines the prefix (here: 90); The ID from the dialog (= acronym) is assigned.
Section [comports]	
com1=0 com2=MSS com3=BAR Com3=LEGIC Com4=RFLESER	Assignment of serial interfaces to connected devices: MSS – machine interface BAR, LEGIC, RFLESER – various reading devices
Section [MSS-INIT]	
MSS_DIALOG=10	If the terminal is switched off longer than 15 minutes, a dialog is displayed on terminal restart. The user must then decide whether the counter pulses, which were recorded when the terminal was closed, are posted or discarded. The dialog closes automatically with "Yes" after an entered time has elapsed; in this case the counting impulses are posted. This value configures the time in seconds the dialog is open. If the terminal is switched off for less than 15 minutes, no dialog is opened; the counting pulses recorded in the switch-off phase are accepted and posted without confirmation. Please note: The value can also be configured in hytnrcfg.ini. Entries in the hytnrcfg.ini file take priority.

Entry	Comment
MSS_FILEAGE_MIN=5 MSS_FILEAGE_OVERTIME=delete	Additional configuration for MSS_DIALOG If the backup file for counting pulses of the terminal is older than 5 minutes, no dialog is opened and the back up file deleted. Quantities recorded at the time when the terminal was closed are not used/posted. Please note: The value can also be configured in hytnrcfg.ini. Entries in the hytnrcfg.ini file take priority.
Section [ext. software]	
Button=Editor WindowName=Editor SearchParts=On	Configuration of the button in the top line: A previously started program can be brought to the foreground at the push of a button. Button: button caption WindowName: Name of the program (e.g. from the taskbar). SearchParts=On: Parts of WindowName are sufficient SearchParts=Off: WindowName must be entered completely. The option "SearchParts=On" is recommended for programs such as MSWord that change the title bar subject to the document that is currently being loaded.
ProgFileName=c:\Programme\win ncmd\Wincmd32.exe	The program that is started if the program mentioned above cannot be called to the foreground.
AutoStart=on	This option starts the program (ProgFileName) when starting the terminal program.
Section [DLL]	
BusDLL=PCC.EXE	PCC.EXE must be entered here if the terminal is configured with the option "operated as HYDRA-MDE terminal". If necessary, the AIP2 enters this value independently upon starting the program.
Section [PDV]	
PDVProtokoll=ON	Enables PDV logging (prot_pdv.txt) (by default=OFF)
PDVIOPDir=c:\IOPSim\	Path for DNC
Section [CAQ]	
;Supported barcodes	
FieldWNRBarcodeOnly=Y	If this entry is set the tool number may only be entered using a scanner.
FieldNestBarcodeOnly=Y	If this entry is set the cavity number may only be entered using a scanner.
FieldNummBarcodeOnly=Y	If this entry is set the number may only be entered using a scanner.
FieldKNRBarcodeOnly=Y	If this entry is set the badge number may only be entered using a scanner.

Entry	Comment
BarcodeWNR=	This field specifies which acronym is entered into the tool number field by the scanner.
BarcodeNest=	This field specifies which acronym is entered into the cavity number field by the scanner.
BarcodeNumm=	This field specifies which acronym is entered into the number field by the scanner.

7.2 PNG – Files / Bitmaps

The use of PNC files is recommended by MPDV. By default PNG files have a size of 24 x 24 px.

7.2.1 File pict.zip

The file "pict.zip" is updated by the installation tool "inst32.exe" while downloading and includes all default PNG files.

The default PNG files can be overwritten in the file pict_cust.zip. Several PNG files have the extension ".small.png" (e.g. aip.small.png). These PNG files are used with a screen resolution of 640x480.

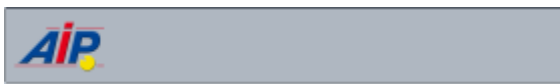
7.2.2 File pict_cust.zip

The file "pict_cust.zip" is loaded from the server directory (e.g. \<serverDir>\1\custom) when starting the program (as is the case for the hycust.mld).

Customized PNG files may be stored in this file and loaded by the AIP2 terminal. Default PNG files may also be "overwritten".

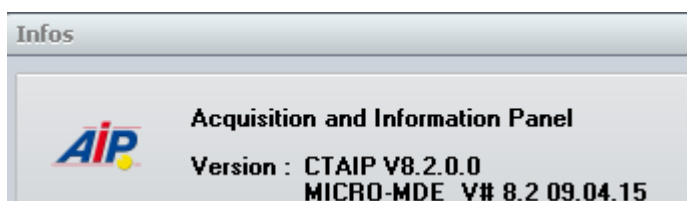
Please note: file sizes are not adjusted.

Customize header



The AIP icon displayed in the header can be replaced by storing a separate AIP.png file in the pict_cust.zip file.

This AIP icon will also be replaced in the "About" dialog.

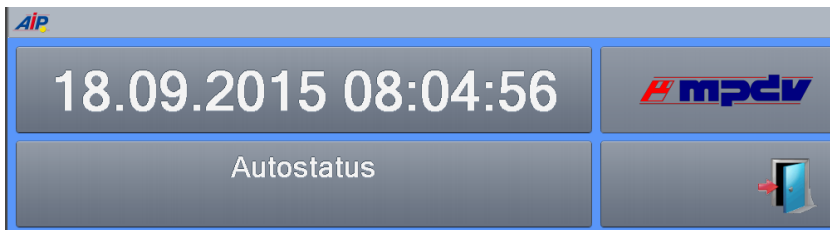


Customize footer



The MPDV icon displayed in the footer can be replaced by storing a separate company.png file in the pict_cust.zip file.

Customize PZE dialog



The MPDV icon displayed in the PZE dialog can be replaced by storing a separate pze_mpdv.png file in the pict_cust.zip file. In case the PZE terminal is operated with a screen resolution of 640x480, a customized pze_mpdv.small.png file has to be integrated in the pict_cust.zip file.

7.3 Multilingualism (*.mld files)

The below-mentioned files are required for the translation of the application. The table shows the priorities for the translation, ownership and the relevant storage locations.

Priority	File	Server directory	Owner	Description
1	ctaipkd.mld	./ctnet/win/aip2/custom	Customer	Customer-specific translations
2	hycust.mld	./1/custom/	MPDV	Customized translations by MPDV. The file is used by the AIP2, CTAIP and CTWIN terminal.
3	ctaip.mld	./ctnet/win/aip2	MPDV	Standard translation file

8 AIP2 - Central Configuration File hytnrcfg.ini

You can use this file as central place to store different configurations for all or for separate terminals.

For each section, a general version is available

[section 0].

The entries included in this section can be overwritten by entries in a terminal-specific section

[section <TNR-USER>]

<TNR-USER> = UserNo = terminal number + 2000 e.g. 2010,2101,..) for exactly one terminal/UserNo



The file hytnrcfg.ini is loaded from the server on every terminal start.

Section / Entry	Comment
	→→→
[Tnr Konfiguration 0]	
FollowExternStatus=on	Transfer of machine statuses when reloading machine list. Useful if status change is set by PDM or another terminal
[Terminal->Installation 0]	
InstallFonts=on	If set to "off", fonts are not installed during restart. ON=DEFAULT
OnlyInstallFontsAfterDownload=false	If "InstallFonts=on": If true, then fonts are only installed directly after a download. If false, then fonts are installed every time the terminal is restarted. (false = DEFAULT)
[Terminal->USR 0]	

Section / Entry	Comment
AttachedApplication=First	Displaying documents of OP info: With this configuration, the system first checks whether or not an application is linked in Windows that matches the file extension of the document. This application is then used to display the document. If no link is available, the viewers configured in ctaip.ini (→ [ext. software]) and internal viewers are used. If an extension is completely unknown, the system tries to display the document as text . Different settings are possible: First → search for linked application first AfterUserViewer → If a UserViewer is configured, this one overrides the linked application (also applies to ExcelViewer, WordViewer and PowerpointViewer) Last → Only if no ctaip.ini assignment is found for the file extension, then the system searches for a linked application (default). Off → The system does not search for a linked application.
HTTPBrowser=standard	Display of documents (via OP info): If documents are configured with a path of schema "http", the file is not downloaded to the terminal, but the link is transferred to a browser. The default browser for the terminal is htmview3.exe, as this one can be operated by touchscreen. If this entry is set, the default browser configured in Windows is used.
SupressErrorMessage=70012	Suppress message "material is not planned"
MSS_DIALOG=10	If the terminal is switched off longer than 15 minutes, a dialog is displayed on terminal restart. The user must then decide whether the counter pulses, which were recorded when the terminal was closed, are posted or discarded. After a configurable period of time, the dialog closes automatically with "Yes" (Yes, posting of pulses). This value configures the time in seconds the dialog is open.
MSS_FILEAGE_MIN=5 MSS_FILEAGE_OVERTIME=delete	If the backup file for counter pulses on the terminal is older than 5 minutes, then no dialog is opened and the backup file deleted. Quantities recorded at the time when the terminal was closed are not used/posted.
[SignatureRecording->User 0]	
ManualBadgeInput=true	This configuration specifies whether or not the field <i>User</i> can be edited on the terminal (by default: no editing) true → activates keyboard input for field <i>User</i> on the terminal

Section / Entry	Comment
Transparency=255	The display of the signature dialog can also be transparent. 255 → Signature dialog is 0 % transparent (not transparent) 1 → Signature dialog is 99% transparent (maximum transparency) (Default = 155)
ShowPosition=TR	You can change the place of the signature dialog: TL Top – Left TM Top – Middle TR Top – Right ML Middle – Left MM Middle – Middle (Default) MR Middle – Right BL Bottom – Left BM Bottom – Middle BR Bottom – Right
USE_SERVICE_ACCOUNT=1	0 (default) SSO: ServiceAccount is not used (requirement: the terminal must be started with the domain "user" (SSO)). Note: ServiceAccount=1 can only be used if all users are in the "root" domain. SubDomain users are not supported.
SIGNATURE_1_USER_TYPE=REPORTING_USER_READONLY	REPORTING_USER_READONLY The user identification using the Windows user is activated. The Windows user is then preassigned in field <i>User</i>. The <i>User</i> field is read-only. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible. REPORTING_USER_CHANGEABLE The user identification using the Windows user is activated. The Windows user is then preassigned in field <i>User</i>. The <i>User</i> field can be edited. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible.

Section / Entry	Comment
SIGNATURE_1_LOGON_TYPE=HYDRA	"" / Not set / "EMPTY" There is also an alternative login procedure. HYDRA The user identification using the Windows user is locked. You must enter a user for identification. Requirement: All reporting persons must have been created as users and the option "SSO" must not be enabled for the users. Otherwise, successful authentication is not possible. ACTIVEDIRECTORY The TAB for the user identification via <i>User</i> is locked. The Windows user must be used for identification purposes. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible. MIXED_BUT_UNIQUE The setting of option "SSO" specifies whether the user login or the Windows login is used. "SSO" enabled → Windows only "SSO" disabled → user only
SIGNATURE_2_LOGON_TYPE=HYDRA	Identical to SIGNATURE_1_LOGON_TYPE (see above)
ExtendedSignatureRecording=true	Used for signatures on the terminal with quality data collection.
[MDE/Blade Configuration 0]	
CONVERT-TO-ANSI- FILE=<list1 list2>	Configuration of the files that are provided from the AIP to the MDEB2 blade in ANSI format when a combined operation is available. The following lists are transferred by default if the entry is not available. counters.lst schicht.lst mnr.lst mstat.lst anr.lst pnr.lst If you want to transfer further lists, you must specify the standard lists and the additional lists.

8.1 Layout configuration

Entry	Comment
Section and/or [terminal configuration 0] [terminal configuration 2XXX];	(general configuration) (2XXX terminal-specific configuration)

Entry	Comment
AUTO-CONFIRM-UHR-ERROR-MESSAGE=TRUE	This setting specifies that in case of an error that occurred reading the clock (e.g. when activated after standby mode), the time is transferred without confirmation dialog and the terminal time is later synchronized with the server time via PDM command.
SUPPRESS-MAXIMUM-NUMBER-OF-MACHINES-WARNING=ON	Suppresses the warning after restart of terminal if more than 32 machines are assigned to the terminal (static/dynamic). (Default=OFF)
CalcTargetYieldSinceLogon=2	CalcTargetYieldSinceLogon=1 The duration is calculated from the total runtime since login (all statuses) minus the configured shift breaks. CalcTargetYieldSinceLogon=2 The duration is calculated from the total runtime since login (all statuses). Defined breaks are not used and are not deducted. (default value)
Section [QRD-PRINTER->TICKET 0] ;(general configuration) [QRD-PRINTER->TICKET 2xxx] ;(2XXX configuration for a specific terminal)	
COMPLETE-ABSENCE-OF-LOCAL-MNR-DATA-FOR-EVENT=< Events >	Reloads the machine row for the configured <Events> , if it is not available locally ➔ This configuration might be required for a group workplace without machine assignment.
COMPLETE-ABSENCE-OF-LOCAL-ANR-DATA-FOR-EVENT=< Events >	Reloads the order row for the configured <Events> , if it is not available locally ➔ This option has been implemented to access order data in the master data, e.g. when logging on orders.
COMPLETE-...-EVENT=< Events > COMPLETE-...-EVENT=#ALL# COMPLETE-...-EVENT=A_AN A_P_AN	Explanation on the configuration of <Events> ➔ Using <#ALL#> the row (ANR/MNR) that is not available is reloaded for any event. ➔ <A_AN A_P_AN> restricts reloading of information to specified events. The ID <DLGFAM> is preferred to the ID <DLG> in order to identify the <Event>.
Section [AIP2 Initialization 0]	
XML-GUI=OFF	Disables the new AIP2 design and uses the AIP 8.1 design.
CTWIN-STYLE=ON	Activates the GUI that is similar to CTWIN on the AIP2. The two button bars are shown below the two lists just like on the AIP 8.1.
CTWIN-BUTTON-LAYOUT=ON	If the option CTWIN-STYLE=ON is additionally set, the two button bars are displayed at the bottom of the screen.
AUTOMATIC-CHANGE-TO-START-DISPLAY=30	As of AIP 8.2.2.28: If this option is set, the display automatically changes to the main view after the configured time if no other interaction was performed in the meantime. The changing display is configured via the option <i>Show machine/OP</i> in the <i>Terminal configuration</i> , tab <i>MF functions</i> . List:

Entry	Comment
	○ Change from the detail views or function menus (operation, person, resource, etc.) ○ Change to the main view with "tiles" • Icons: ○ Change from the detail views or function menus or from the main view with tiles ○ Change to the icon view of workplaces The configuration is specified in seconds. Do not specify less than 10 seconds. An automatic change to the start screen is not made if an input dialog is open.